1. What is the lowest level of knowledge, representing raw inputs like numbers, characters, or images?

- A) Information
- B) Metadata
- C) Data
- D) Schema

2. Which of the following best describes "Information"?

- A) Raw input that has no significance by itself
- B) Processed or arranged data that makes a meaningful output
- C) Physical storage addresses on secondary storage devices
- D) Unorganized recordings of observations

3. Which of the following is a key drawback of using a traditional file-based system?

- A) Centralized data management
- B) Data redundancy and inconsistency
- C) Automatic enforcement of global integrity constraints
- D) High physical and logical data independence

4. What is the role of a Database Management System (DBMS)?

- A) It serves as an intermediate layer between the database and the programs that access the data.
- B) It replaces all application programs with manual paper processing.
- C) It operates strictly as a sequential tape storage device.
- D) It eliminates the need for data modeling.

5. Which of the following is listed as a core function/service of a DBMS?

- A) Manual data duplication
- B) Concurrency control and recovery
- C) Eliminating the need for operating systems
- D) Enforcing permanent application incompatibility

6. What are some of the known disadvantages of database systems?

- A) High redundancy and total data isolation
- B) Need for expertise to use, expense of the DBMS, and potential incompatibility
- C) Inability to share data or apply security restrictions
- D) Complete absence of backup and recovery mechanisms

7. In the three-level database architecture, which level describes the physical storage structure of data and uses a physical data model?

- A) External level
- B) Conceptual level
- C) Internal level
- D) User interface level

8. Which level in the three-schema architecture describes the structure of the whole database for a group of users and hides internal storage details?

- A) Physical level
- B) Conceptual level
- C) External level
- D) Hardware level

9. What does the external level of database architecture focus on?

- A) The entire enterprise-wide database structure
- B) The data as viewed by individual end users (user views / external schemas)
- C) Sector-level addresses on secondary storage devices
- D) Operating system file allocation tables

10. Who is the person with central control over the database system, responsible for schema definition, storage structure modification, and authorization granting?

- A) Application Programmer
- B) End User
- C) Database Administrator (DBA)
- D) Web Commerce Developer

11. What are the duties of Application Programmers in a database environment?

- A) Designing and implementing application programs, data-entry screens, and reports
- B) Defining organization-wide conceptual schemas and storage methods
- C) Performing routine database backups and monitoring free disk space
- D) Managing hardware upgrades for multi-terabyte data warehouses

12. Which of the following is an advantage of a database system over a file-based system regarding data management?

- A) Data can be shared, and redundancy can be reduced
- B) Programmers must rewrite code for every new task
- C) Integrity constraints remain buried inside application logic

- D) Inconsistency across independent file formats is maximized

13. What is the main daily responsibility of a database designer?

- **A)** Handling routine business operations and processing customer orders
- **B)** Building efficient database frameworks utilizing tools such as SQL and ER diagrams
- **C)** Replacing hard drives and performing physical server upkeep
- **D)** Developing batch application programs for magnetic tape systems

14. At the highest level, how can data at the physical level be viewed?

- A) In the form of files
- B) As individual end-user external views
- C) As raw bits with physical addresses on secondary storage
- D) As relational conceptual tables

15. What does the internal schema define in database architecture?

- A) User views and external schemas
- B) The various stored data types using a physical data model
- C) The overall enterprise conceptual model
- D) Network transmission protocols

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1. Which stage of the database development life cycle involves high-level project planning and understanding how the enterprise works?

- A) Implementation
- B) Database Planning
- C) Physical Modeling
- D) Data Conversion and Loading

2. What is the primary purpose of the "Systems Definition" stage?

- A) Writing production code and executing test queries
- B) Specifying scope, boundaries, and how the system operates from different perspectives
- C) Translating the database model into a physical DBMS implementation
- D) Transferring existing data from an old system to a new system

3. Which database design stage focuses on developing a model on paper without any physical implementations or specific DBMS considerations?

- A) Physical model
- B) Logical model

- C) Implementation phase

5. During which implementation phase are errors analyzed, and system performance, robustness, and adaptability tested?

- A) Requirement Analysis
- B) Logical Design
- C) Testing
- D) System Definition

6. In an ER diagram, what do rectangular boxes traditionally represent?

- A) Attributes
- B) Relationship sets
- C) Entity sets
- D) Constraints

7. How is a weak entity distinguished from a strong entity in an ER diagram?

- A) It is enclosed in a dashed ellipse.
- B) It is enclosed in a doubly outlined rectangle (double rectangle).
- C) It is connected via a diamond with a primary key underline.
- D) It has no attributes of its own.

8. What is a weak entity?

- A) An entity that exists independently of all other entities.
- B) An entity whose instances cannot be defined solely by its own attributes because it lacks sufficient attributes to form a primary key.
- C) An entity that only participates in ternary relationships.
- D) An entity that has no relationships with any other table.

9. What is a simple attribute?

- A) An attribute composed of multiple sub-parts like first and last name.
- B) An attribute whose value is atomic and cannot be further divided.
- C) An attribute derived dynamically from birthdate.
- D) An attribute that holds multiple values simultaneously.

10. Which type of attribute has sub-parts that spring from it, such as a name breaking down into first, middle, and last names?

- A) Composite Attribute
- B) Derived Attribute
- C) Multi-valued Attribute
- D) Simple Attribute

11. How is a derived attribute represented in an ER diagram?

- A) Solid line ellipse
- B) Double-lined ellipse
- C) Dashed ellipse
- D) Doubly outlined rectangle

12. What does a multi-valued attribute represent?

- A) An attribute calculated from another field.
- B) More than one attribute value denoted for a single entity instance, such as multiple phone numbers.
- C) An attribute shared across three separate entities.
- D) A primary key combined with a foreign key.

13. What are the key characteristics of a Primary Key?

- A) It can contain null values for optional instances.
- B) It can change its value over time as records update.
- C) It uniquely identifies an instance, must be unique, must always have a value, and can consist of one or more attributes.
- D) It is only used to connect two independent file systems.

14. What is a Candidate Key?

- A) A key that always contains null values.
- B) Any of the multiple possible unique identifiers when more than one identifier exists.
- C) An attribute that is calculated from the current date.
- D) A foreign key pointing to a dormitory table.

15. What does a Foreign Key establish in a database?

- A) It references a related table through the primary key of that related table.
- B) It creates a secondary derived attribute.
- C) It defines the physical storage blocks on secondary memory.
- D) It enforces unary recursive relationships exclusively.

16. What does the "Degree of a Relationship" refer to?

- A) The number of entity types that participate in a relationship.
- B) The maximum number of tuples linked in a cardinality ratio.
- C) The number of simple attributes attached to an entity.
- D) The depth of a composite attribute structure.

17. What is a Unary (Recursive) Relationship?

- A) A relationship between three distinct entity types.
- B) A relationship in which the same entity participates more than once (e.g., an employee supervising another employee).
- C) A relationship that connects a customer to a loan.
- D) A relationship with a strict 1:1 mapping constraint only.

18. What does a Ternary Relationship set have a degree of?

- A) Degree 1
- B) Degree 2
- C) Degree 3
- D) Degree 4

19. What is Mapping Cardinality?

- A) The total number of tables in a database schema.
- B) How many instances of one entity will or must be connected to a single instance from the other entities.
- C) The minimum physical storage size required for an entity set.
- D) The rule governing derived attribute calculations.

20. What does a double line between an entity and a relationship in an ER diagram indicate?

- A) Partial participation
- B) Total participation (mandatory participation)
- C) A multi-valued attribute connection
- D) A ternary relationship link

21. In the context of participation constraints, what does "Partial Participation" mean?

- A) Every entity instance must participate in the relationship instance.
- B) Only some entity instances participate in the relationship (e.g., only some employees manage departments).
- C) The entity is a weak entity that cannot exist without an owner.
- D) The entity contains null values in its primary key.

23. What is the primary function of an Entity-Relationship (ER) model?

- A) To specify physical disk block allocation and sector sizes
- B) To develop a conceptual design and provide a simple view of data for a specified system
- C) To compile high-level DML statements into executable binary files
- D) To execute automated database backup and recovery operations

24. In an ER diagram, how is an entity set traditionally represented?

- A) Rectangle
- B) Diamond
- C) Ellipse
- D) Dashed line

25. Which of the following best defines a weak entity?

- A) An entity that has multiple key attributes of its own
- B) An entity that depends on another entity and does not contain any key attribute of its own
- C) An entity that is always connected via a 1:1 mapping constraint
- D) An entity that contains exclusively derived attributes

26. How is a weak entity represented in an ER diagram?

- A) Single-line rectangle
- B) Doubly outlined box or rectangle
- C) Diamond with underlined text
- D) Dashed ellipse

27. What geometric shape is used to represent an attribute in an ER diagram?

- A) Rectangle
- B) Diamond
- C) Ellipse
- D) Triangle

28. What is a composite attribute?

- A) An attribute that consists of a single atomic value like salary
- B) An attribute composed of many other attributes (not atomic)
- C) An attribute whose value is calculated from a Date of Birth
- D) An attribute that holds multiple phone numbers

29. How is a multi-valued attribute represented in an ER diagram?

- A) Ellipse with underlined text
- B) Dashed ellipse
- C) Doubly lined ellipse
- D) Rectangle with a double border

30. Which type of attribute has its value computed or derived from another stored attribute, such as calculating age from Date of Birth?

- A) Simple attribute
- B) Composite attribute
- C) Derived attribute
- D) Multi-valued attribute

31. What does a dashed ellipse represent in an ER diagram?

- A) Key attribute
- B) Derived attribute
- C) Multi-valued attribute
- D) Total participation

32. What is the degree of a relationship in an ER model?

- A) The number of entity types that participate in a relationship
- B) The maximum number of tuples linked in a cardinality ratio
- C) The total number of attributes possessed by an entity
- D) The depth of composite attribute nesting

33. Which relationship degree involves instances of three different entity types?

- A) Unary
- B) Binary
- C) Ternary
- D) Recursive

34. What is a unary (recursive) relationship?

- A) A relationship between two separate entity types
- B) A relationship between two instances of the same entity type
- C) A relationship among three distinct entity types
- D) A relationship with a 1:M cardinality ratio between departments and employees

35. What does mapping cardinality express in a database model?

- A) The physical byte size of stored records
- B) The number of entities to which another entity can be related via a relationship set
- C) The number of simple attributes attached to a weak entity
- D) The sequence of records stored on magnetic tape

36. What are the properties of a Primary Key?

- A) It must be unique, must always have a value (not null), and its value should not change over time
- B) It can contain null values for optional records
- C) It can be duplicated across multiple tuples in a table

- D) It is restricted to exactly one attribute per table

37. How is a candidate key defined in relation to a primary key?

- A) A candidate key is any foreign key pointing to a parent relation
- B) An attribute or set of attributes that can uniquely identify a tuple, from which a primary key is chosen
- C) A composite key that must span across three or more tables
- D) A super key that contains redundant non-minimal attributes

38. What is a super key?

- A) Any attribute set that can uniquely identify a tuple, acting as a superset of a candidate key
- B) A key that always contains null values
- C) An alternate term for a derived attribute
- D) A constraint restricting participation to total only

39. What is a foreign key?

- A) A primary key that uniquely identifies tuples in its own table
- B) A column in a table used to point to the primary key of another table
- C) An attribute represented by a dashed ellipse
- D) A multi-valued attribute holding phone numbers